LECTURE 3: RANDOM VARIABLES AND FUNCTIONS THAT DESCRIBE THEM

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to `elre@princeton.edu`, thanks in advance!

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Recall that we call a pair $(S, \mathcal{F})$ of an arbitrary set S with a σ-algebra $\mathcal{F}$ on it a *measurable space*, meaning that one can define a measure on it. If μ is a measure defined on the pair, then call $(S, \mathcal{F}, \mu)$ is a *measure space* (or a *probability space* if μ is a probability measure).

Definition 3.1 (Random variable)**.** A function $f : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ is called a **measurable function** if for any $A \in \mathcal{S}$, the preimage $f^{-1}(A) = \{x \in \Omega : f(x) \in A\}$ is also measurable. A **random variable** is a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (S, \mathcal{S})$.

The idea behind this definition is as follows: If we want to know the probability that a random variable takes a certain value, or is in a certain (measurable) range, we need the probability function to be well-defined on the preimage of the range or value of interest. Consider an example: We select a number from $\Omega = \{1, 2, \dots, 100\}$ uniformly at random. We are interested what is the probability that this number (mod 5) gives 3. We naturally have a function $X : \Omega \to \{0, 1, 2, 3, 4\}$ ($\omega \mapsto w \mod 5$). The question above asks for

$$\mathbb{P}(X^{-1}(3)) = \mathbb{P}(\{w : X(w) = 3\}) =: \mathbb{P}(X = 3).$$

Remark 3.2.
(1) Notation: We write $\mathbb{P}(X \in A)$ to denote $\mathbb{P}(\{w : X(w) \in A\})$. The dependence of a random variable on the sample space will frequently be omitted when it is clear. However, it is important to remember that a random variable X is a function defined on some underlying space Ω (i.e., its value depends on the "state of the world" $\omega \in \Omega$.)
(2) If the target space $(S, \mathcal{S})$ is unspecified, it is typically assumed that it is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e., X is a real-valued random variable.

The following lemma lists some useful properties of measurable functions that can be used to claim measurability of newly constructed functions.

Lemma 3.3. *Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ and $Y : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be measurable functions.*
(1) (Generated σ-algebras). *If $\mathcal{S} = \sigma(\mathcal{A})$, then X is measurable if and only if $X^{-1}(A)$ is measurable for any $A \in \mathcal{A}$.*
(2) (Composition of RVs). *If $Z : (S, \mathcal{S}) \to (E, \mathcal{E})$ is measurable, then $Z \circ X$ is also measurable.*
(3) (Pointwise maxima/minima). *If $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then the pointwise maximum* $\max\{X, Y\}$ *and minimum* $\min\{X, Y\}$ *functions are measurable. In particular, the*

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positive and negative parts $X_+ = \max\{X, 0\}$ *and* $X_- = -\min\{X, 0\}$ *of* $X = X_+ - X_-$ *are also measurable.*

(4) (Sums, differences, products and ratios). *If* $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$*, then the functions (defined pointwise)* $X + Y$*,* $X - Y$*,* XY *and* X/Y *(provided* $Y \neq 0$ *everywhere) are measurable.*

(5) (Pointwise limits). *If* $X_n : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$*,* $n \geq 1$*, is a sequence of real-valued measurable function, then the functions (defined pointwise)* $\sup_n X_n$*,* $\inf_n X_n$*,* $\limsup_n X_n$*,* $\liminf_n X_n$*, and* $\lim X_n$ *(when the limit exists, i.e.,* $\limsup = \liminf$*) are also measurable.*

(6) (Continuous/monotone functions). *Let* $f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ *be a real-valued function. If* f *is left-continuous, right continuous, or monotone, then* f *is measurable.*

Most of these properties are not hard to check; some of these are homework exercises. For example, for property (1), consider $\mathcal{B} := \{B \subset \mathcal{S} : X^{-1}(B) \text{ is measurable}\}$. If the preimages of the generator, $X^{-1}(\mathcal{A})$, are measurable, then it can be checked from the definition that $\mathcal{B}$ is a σ-algebra that contains $\mathcal{A}$, and hence it contains $\sigma(\mathcal{A})$.

Three objects associated with a RV: σ-algebra $\sigma(X)$, the law of X, the CDF of X

(1) The law of X is a probability measure on the image space (typically, on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$), denoted as $\mathcal{P}_X$. It allows us to talk about random variables without references to the original space Ω and only focus on the probabilities of the observed values.

Definition 3.4 (Random variables define new probability measures)**.** Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be a random variable. The **law of** X is the probability measure $\mathcal{P}_X$ on the image space induced by the random variable X, defined as $\mathcal{P}_X(A) := \mathbb{P}(X \in A)$ for any $A \in \mathcal{S}$.

Exercise: check that the function $\mathcal{P}_X$ indeed defines a probability measure.

(2) The generated σ-algebra $\sigma(X)$ contains exactly those events A for which we can say whether ω is in A or not, based on the value of $X(\omega)$.

Definition 3.5 (Random variables define new σ-algebras)**.** Given a random variable X, the σ-algebra **generated by** X, denoted by $\sigma(X)$, is the *minimal* σ-algebra such that $X(\omega)$ is a measurable function.

Exercise: Consider an experiment where an infinite sequence of coin tosses is observed. The outcome space has elements $\omega = (\omega_1, \omega_2, \ldots)$, where each $\omega_i \in \{H, T\}$. Let's identify $H = 1$ and $T = 0$. Note that any infinite sequence of zeros and ones is possible, and thus the space can be identified with the segment $[0, 1]$ by binary expansions. Let $X(\omega) := \omega_n$ (the result of the nth toss). What is the σ-algebra generated by X, $\sigma(X)$? Think this through. (The image of X is $\{0, 1\}$. $X^{-1}(\{0\}) =$ all sequences with 0 in the nth position. $\ldots$)

(3) If a random variable maps to $\mathbb{R}$, the cumulative distribution function F_X essentially presents a more convenient representation for its law, focusing only on the set of generators for a Borel sigma-algebra.

Definition 3.6. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **cumulative distribution function (CDF)** of X is a function $F_X : \mathbb{R} \to [0,1]$ defined as $F_X(t) := \mathbb{P}(X \leq t)$.

Theorem 3.7. *Two random variables X, Y have the same CDF if and only if they have the same law.*

Proof. Consider the set of all intervals $\Pi := \{(-\infty, b] : b \in \mathbb{R}\}$. This is a π-system (check this!) of subsets of $\mathbb{R}$ that generates $\mathcal{B}(\mathbb{R})$. By definition, the law of X satisfies $\mathcal{P}_X((-\infty, b]) = \mathbb{P}(X \leq b) = F_X(b)$. Hence, if any other law $\mathcal{P}_Y$ has the same CDF, then $\mathcal{P}_X = \mathcal{P}$ on $\mathcal{B}(\mathbb{R})$ by the corollary of the Dynkin π-λ theorem from Lecture 2. □

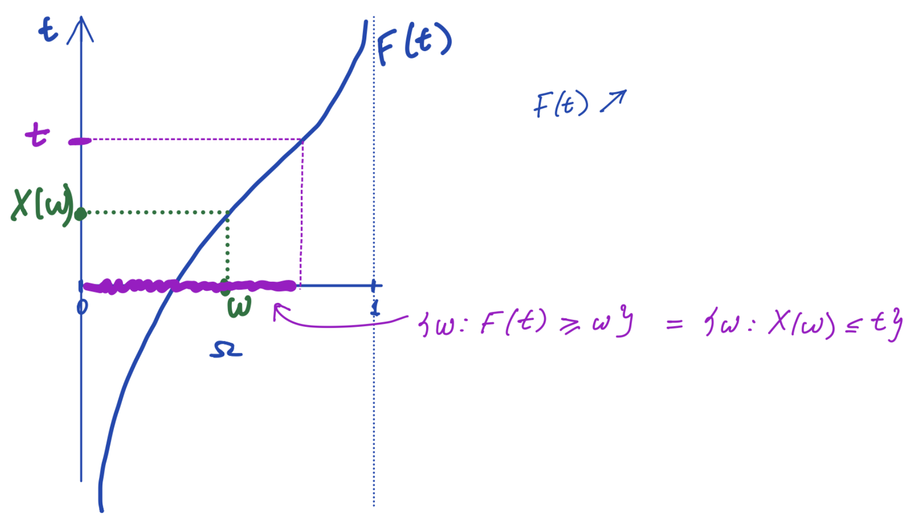
Theorem 3.8. *Any CDF $F : \mathbb{R} \to [0,1]$ satisfies the following properties:*

(1) *F is non-decreasing*
(2) *F is right-continuous*
(3) $\lim_{t\to-\infty} F(t) = 0$ *and* $\lim_{t\to\infty} F(t) = 1$.

If a function F satisfies (1)–(3), then there exists a probability space and a random variable X defined on this space such that its CDF F_X coincides with F.

Proof of Theorem 3.8. We can verify the properties (1)–(3) for any CDF using monotonicity, continuity from above and from below of the law $\mathcal{P}_X$ of the constructed random variable X with the same CDF F (exercise!).

Consider any F satisfying the properties (1)–(3). Let $\Omega = (0,1)$, and consider the probability space $((0,1), \mathcal{B}(0,1), \mathbb{P})$, where $\mathbb{P}$ is the uniform measure (i.e., $\mathbb{P}$ is the Lebesgue measure on the unit interval). Define the function $X(\omega) := \inf\{y : F(y) \geq \omega\}$ on the probability space. Note that X is well-defined due to (3). Furthermore, properties (1) and (2) imply that $\{\omega : X(\omega) \leq t\} = \{\omega : F(t) \geq \omega\}$; see the following figure:



Therefore,

$$F_X(t) = \mathbb{P}\{X \leq t\} = \mathbb{P}\{\omega : X(\omega) \leq t\} \overset{\text{(i)}}{=} \mathbb{P}\{\omega : \omega \leq F(t)\} \overset{\text{(ii)}}{=} \mathbb{P}((0, F(t)]) \overset{\text{(iii)}}{=} F(t).$$

Here, (i) is by the identity in the purple box above, (ii) is simply notation, and (iii) is by the definition of the uniform probability measure on $(0,1)$.

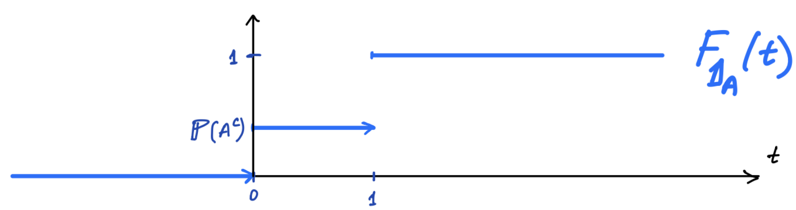
Finally, this shows that X is measurable (since F is measurable as a non-decreasing function). Hence, X is a random variable with the same CDF as F. $\square$

Remark 3.9. One can check that a CDF can have at most countably many points of discontinuity.

Basic example: indicators and simple random variables, approximation by simple random variables

Some fundamental random variables are the following:

- *Indicator (Dirac) random variables:* fix $A \in \mathcal{F}$, and consider the indicator random variable of the event A:
$$\mathbb{1}_A(\omega) = \begin{cases} 1 \text{ if } \omega \in A, \\ 0 \text{ otherwise.} \end{cases}$$
Let's discuss this. (1) $\mathbb{1}_A$ is a random variable since, for example, $X^{-1}(B) = A$ if $1 \in B$ and $0 \notin B$; the other cases include the preimages $\emptyset, A^c, \Omega$, which are all measurable by definition of the σ-algebra $\mathcal{F}$. (2) $\sigma(X) = \{\emptyset, A, A^c, \Omega\}$. (3) The law of X is given by $\mathcal{P}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(A)$ if $1 \in B$ and $\mathbb{P}(A^c)$ otherwise for any $B \in \mathcal{B}(\mathbb{R})$).
The CDF of an indicator (Dirac) random variable looks like the following:

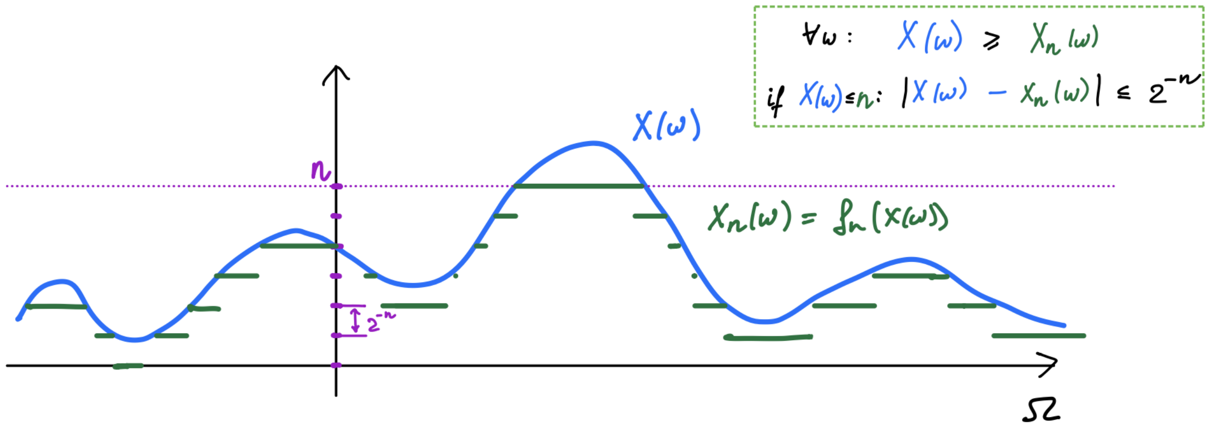


- *Simple random variables:* for any finite sequence of events $A_1, \ldots, A_N \in \mathcal{F}$ and non-random numbers $c_1, \ldots, c_N$, we can define a simple random variable by the following finite sum:
$$X(\omega) := \sum_{n=1}^{N} c_n \mathbb{1}_{A_n}(\omega).$$

Theorem 3.10 (Monotone approximation by simple functions)**.** *For any random variable $X(\omega)$, there exists a monotone increasing sequence of simple random variables $X_n(\omega)$ such that $X_1(\omega) \leq X_2(\omega) \leq \ldots$ and $X_n(\omega) \to X(\omega)$ as $n \to \infty$ for any $\omega \in \Omega$.*

If $X \geq 0$ is nonnegative a.s., then the sequence is pointwise monotonically increasing.

Proof. Define functions $f_n : \mathbb{R} \to \mathbb{R}$ by
$$f_n(x) := n\mathbb{1}_{\{x>n\}}(x) + \sum_{k=0}^{n2^n-1} k2^{-n}\mathbb{1}_{(k2^{-n},(k+1)2^{-n}]}(x).$$
If $X \geq 0$, then $X_n := f_n(X)$ is such that $X \geq X_{n+1} \geq X_n$ and $X(\omega) - X_n(\omega) < 2^{-n}$ if $X(\omega) \leq n$. So, $f_n(x) \to f(x)$ as $n \to \infty$. For arbitrary $X(\omega)$, we represent it as $X(\omega) = X_+(\omega) - X_-(\omega)$ with $X_+(\omega) := \max(X(\omega), 0)$ and $X_-(\omega) := -\min(X(\omega), 0)$, and apply the construction above to the positive and negative parts separately. $\square$



Probability density functions

Some (but not all) random variables also have density functions.

Definition 3.11. A random variable $X(\omega)$ has a **probability density function** $f_X : \mathbb{R} \to \mathbb{R}$ with respect to Lebesgue measure if

$$\mathbb{P}(X \in A) = \int_A f_X(x)\,\mathrm{d}x$$

for all Borel sets $A \in \mathcal{B}(\mathbb{R})$.

What do we mean by "with respect to Lebesgue measure?" How do we integrate over an arbitrary Borel set? Soon, we will also want to integrate arbitrary measurable functions (and random variables). In the next lecture, we will formally define integration with respect to some measure (in particular, we will define the Lebesgue integral with respect to the Lebesgue measure).

For now, note that the definition of a random variable with a probability density function f_X implies the following:

(1) $\int_{\mathbb{R}} f_X(x)\,\mathrm{d}x = 1$.
(2) $f_X(x) \geq 0$ almost everywhere, i.e., for all x except on a set of measure zero.
(3) The CDF F_X of X is given by

$$F_X(t) = \int_{-\infty}^{t} f_X(x)\,\mathrm{d}x.$$

(Compare the properties of integrals with the properties of CDFs. For now, you can think of the integral in a Riemann sense.)